

Perimity — Event & Bulk Visitor Design

Smart Campus Access & Gate Pass Management System Design decisions for events, bulk onboarding, and the gate scan logic.

Guiding Principle

Replace the paper register at the gate. Entry-only. Passwordless (email + OTP) for visitors. Every design choice serves a faster, searchable, forgery-proof version of the guard’s notebook.

Core Model: A Pass Is Not a Person

Two concepts kept separate:

- **Identity** — who you are. One per person, forever, keyed by email.
- **Pass** — permission to enter, for a purpose, for a time window. A person can hold many at once.

A student can simultaneously hold:

- a **DAILY** pass (their standing student pass, no end date), and
- an **EVENT** pass (for a specific programme, valid only for the event dates).

Both active at the same time is normal — like holding a company ID badge and a separate conference lanyard.

Login Model

User	Login method
Visitor	Email + OTP only, no password
Event visitor	Email + OTP only, can re-view/download pass anytime
Student / Faculty / Admin	Email + OTP (passwordless everywhere, per project decision)

A one-time event visitor never creates a password. If they lose the pass email, they go to the site, enter email, get OTP, and see their pass again.

Bulk Onboarding — One Engine for Everything

There is a **single bulk engine**, not separate flows for students vs visitors. The only difference is a `type` column and where the dates come from.

Type	Dates come from	Pass produced
student	none (permanent)	DAILY pass, no end date
visitor	the event's date range	EVENT pass, event dates

Excel columns (minimum)

For an event visitor batch: `name, email, phone, purpose`. Visit dates are **not** per-row — the event's date range applies to the whole batch (e.g. Aug 10–12 set once for all 600).

Validation and confirmation flow

```
Faculty/Admin uploads Excel (e.g. 600 rows for "AI Summit")
  ↓ (fast path – faculty waits ~2 seconds)
Parse + validate every row
  ↓
Show summary: "580 valid, 20 errors"
  ↓
Faculty clicks Confirm
  ↓
Create identities + drop QR/PDF jobs into RabbitMQ
  ↓
Faculty sees "Done – passes are being generated" (faculty can leave)
  ↓ (slow path – background, faculty not waiting)
QR/PDF service processes jobs one by one:
  generate token → QR PNG → PDF → upload to S3 → email pass
  ↓
Each visitor gets: welcome email + QR pass PDF attached
```

Why async: generating 600 QRs + 600 PDFs + 600 emails synchronously would time out the browser and half would fail. RabbitMQ decouples this so the faculty never waits and one failure never blocks the rest.

Handling bad rows

- Process all valid rows; never block the batch for a few bad ones.
- Return a downloadable error report: `row 34: invalid email`, `row 51: duplicate`.
- Faculty fixes only those rows and re-uploads them.

Duplicate / blocklist checks during validation

- Same email twice in the sheet, or already has a pass for this event → skip, flag in error report.
- Email on the campus blocklist → skip automatically, flag in report.

The Mixed-Attendee Problem (600 attendees, 102 already students)

The faculty uploading 600 rows should **not** have to know who is already in the system. The engine decides per row, matched by email:

```
For each row (matched by email):
  | email exists as a student/faculty of ANY campus?
  |   → reuse their identity, issue only an EVENT pass
  |   (no duplicate account created)
  |
  | email is brand new (totally outside)?
  |   → create a lightweight VISITOR identity
  |   (name, email, phone – no roll number, no department)
  |   + issue an EVENT pass
```

So all 600 receive an **event pass**, but only the new people get a new identity. The 102 existing students are recognized by email and linked to their existing identity. No duplicates.

Cross-campus attendees

- **From another campus in our system** (e.g. CDAC Pune student at CDAC Mumbai's event) → recognized by email, event pass scoped to Mumbai's event, their existing identity reused.
- **From totally outside** → brand new lightweight visitor identity + event pass. After the event the pass expires; the identity stays only for audit ("who attended AI Summit 2026").

The email is the universal key in every case.

Which QR Matters During an Event (Option B)

During event days, the **event QR is the one that matters** — scanning it logs entry against the event, giving the organizer a clean attendance list. The student's daily pass still works for normal campus entry outside the event.

The welcome email tells the visitor: "use this QR for the programme."

The two-QR problem and how it's handled (Behavior 2)

A student attending the event has two valid QRs (daily + event). If they scan the **daily** QR out of habit during event days, the system is smart about it:

If this person has an event running today, auto-attribute the entry to that event — even though they scanned the daily QR.

Green light, one scan, no gate friction — and the organizer still gets accurate attendance because the system does the attribution, not the guard.

Gate Scan Logic (Entry-Only)

```
Guard scans QR
↓
Decrypt token → identify person + which pass
↓
Pass valid + active + in date range?
├ NO → RED + reason (expired / revoked / invalid), log denied attempt
└ YES
    ↓
    Is this an EVENT pass?
    ├── YES → log entry against event_id → GREEN "Welcome to [Event]"
    └ NO (daily pass)
        ↓
        Does this person have an event running today? (Behavior 2)
        ├── YES → log entry against that event → GREEN "Welcome to [Event]"
        └ NO → log normal campus entry → GREEN "Welcome"
```

- One scan, instant green — the branching is invisible to the guard.
- Entry-only: no exit scan, no in/out toggle.
- Repeat entries in a day are all logged (a person may enter multiple times; each is a row), matching how a paper register would have multiple lines.
- For a multi-day event, each day's first scan is that day's attendance; a 3-day event = up to 3 entry logs per person, grouped by day in the organizer view.

Organizer Attendance View (the payoff)

A paper register could never produce this — it is the strongest demo moment for the event feature:

AI Summit · Aug 10–12

Registered: 600

Attended Day 1: 543 (90%)

Attended Day 2: 478 (80%)

Never showed: 41

[\[Search attendee\]](#) [\[Export attendance CSV\]](#)

Welcome Email (pass delivery + welcome in one)

Subject: Your gate pass for AI Summit at CDAC Mumbai

Hi [Name],

Welcome to AI Summit on Aug 10–12.

Your entry QR pass is attached – show it at the gate.

Valid: Aug 10–12, Main Gate.

– CDAC Mumbai

One email, QR PDF attached. Welcome + pass together, not two separate emails.

Schema Additions for Events

New `events` table:

Column	Notes
<code>id</code>	PK
<code>campus_id</code>	FK — event belongs to a campus
<code>name</code>	e.g. "AI Summit"
<code>valid_from</code>	event start date
<code>valid_to</code>	event end date
<code>created_by</code>	faculty/admin user id

Additions to `gate_passes` :

Column	Notes
<code>pass_type</code>	<code>DAILY</code> or <code>EVENT</code>
<code>event_id</code>	null for daily passes, set for event passes

So one person can hold:

- pass A: `type=DAILY, event_id=null` (normal student pass)
- pass B: `type=EVENT, event_id=17, valid Aug 10-12` (the programme)

Both active, no conflict.

Decisions Locked

1. One identity per person, many passes — daily and event passes coexist.
2. Bulk engine is shared for students and event visitors; only `type` and date source differ.
3. Mixed attendee batches resolved per-row by email — existing users reused, new people get lightweight visitor identities.
4. Event QR is authoritative during events (Option B).
5. Daily QR scanned during an event auto-attributes to the running event (Behavior 2).
6. Entry-only; every entry logged; multi-day events log one attendance per day.